

Bangladesh University of Business and Technology



LAB REPORT

Experiment No : 02

Course Title : Database Systems Lab

Course Code : CSE 208

Experiment Name: Guided Practice Problems, Individual Practice /
Evaluation, Practice Problem

Submitted By:

Name : Luckman Rafique

ID No : 20255103042

Intake : 55

Section : 02

Program : B.Sc. in CSE

Submitted To:

Name : Md. Shakil Ahmed

(Lecturer)

Dept. of CSE

**Bangladesh University of Business
& Technology**

Date of Submission : 22/09/2026

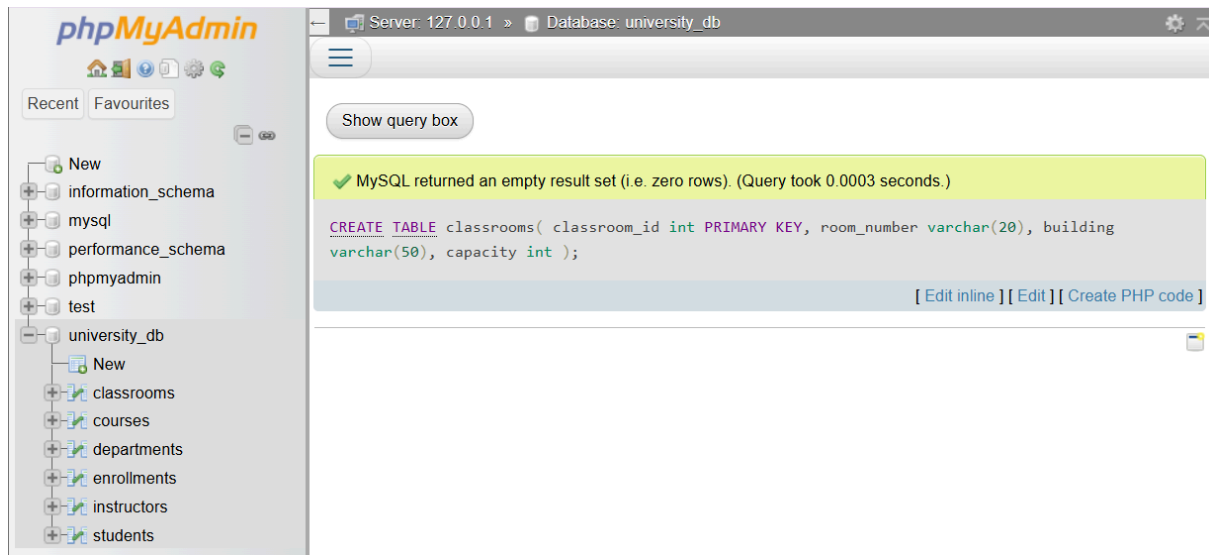
Part (i) — Guided Practice Problems —

Question 1. Create a table named classrooms with classroom_id, room_number, building, and capacity.

Code:

```
CREATE TABLE classrooms(  
    classroom_id int PRIMARY KEY,  
    room_number varchar(20),  
    building varchar(50),  
    capacity int  
);
```

Output:



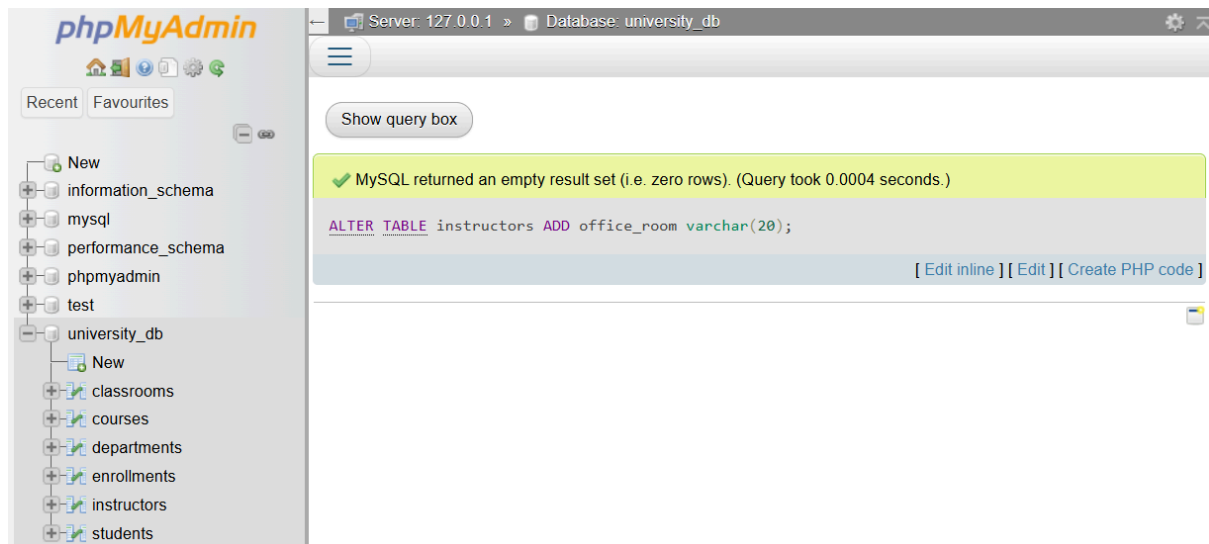
The screenshot displays the phpMyAdmin web interface. On the left sidebar, the 'university_db' database is selected, and the 'classrooms' table is visible under the 'New' section. The main panel shows the results of a SQL query. A green message box indicates: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)'. Below this, the executed SQL query is shown: `CREATE TABLE classrooms( classroom_id int PRIMARY KEY, room_number varchar(20), building varchar(50), capacity int );`. At the bottom right of the query box, there are links: '[Edit inline] [Edit] [Create PHP code]'.

Question 2. Add a column office_room to instructors.

Code:

```
ALTER TABLE instructors  
ADD office_room varchar(20);
```

Output:



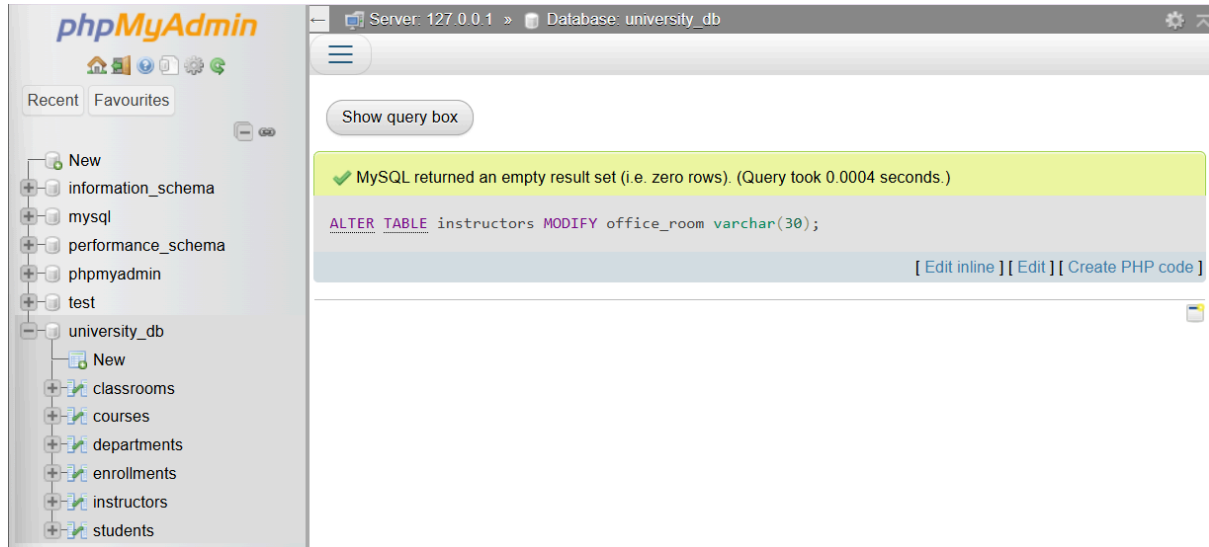
instructor_id	instructor_name	deparment_id	office_room
101	Dr. Rahman	1	NULL
102	Dr. Akif	2	NULL
103	Prof. Dr. Khandakar Nadim Parvez	3	NULL

Question 3. Modify office_room so it can store up to 30 characters.

Code:

```
ALTER TABLE instructors MODIFY office_room varchar(30);
```

Output:



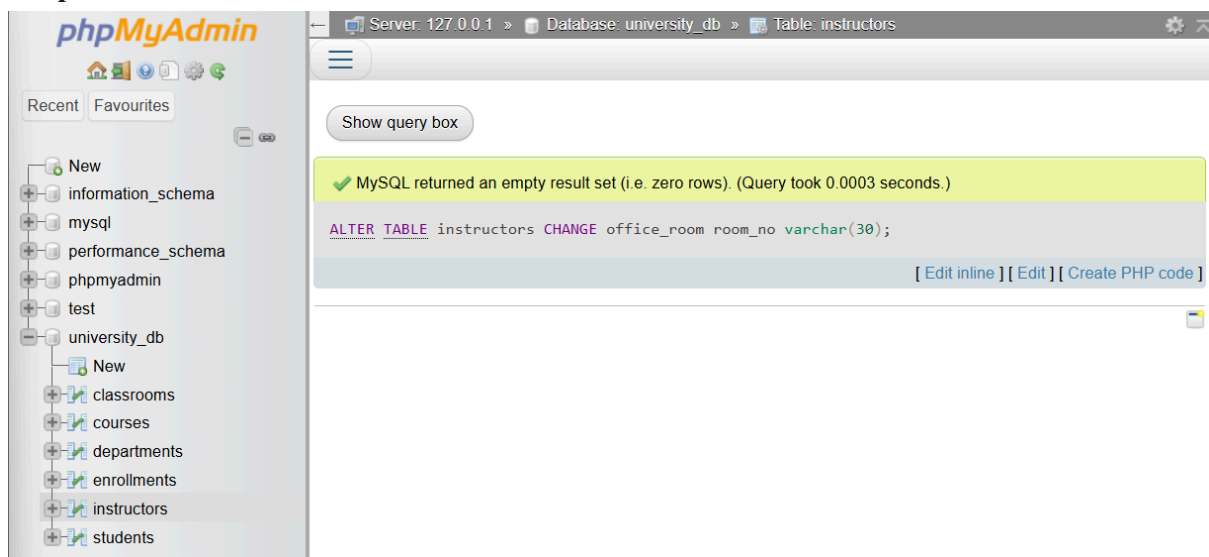
The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, with 'university_db' expanded to show tables like 'classrooms', 'courses', 'departments', 'enrollments', 'instructors', and 'students'. The main panel displays the query 'ALTER TABLE instructors MODIFY office_room varchar(30);' and a success message: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)'. Below the query, there are links for '[Edit inline]', '[Edit]', and '[Create PHP code]'.

Question 4. Rename office_room to room_no.

Code:

```
ALTER TABLE instructors  
CHANGE office_room room_no varchar(30);
```

Output:



The screenshot shows the phpMyAdmin interface. The main panel displays the query 'ALTER TABLE instructors CHANGE office_room room_no varchar(30);' and a success message: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)'. Below the query, there are links for '[Edit inline]', '[Edit]', and '[Create PHP code]'. The breadcrumb at the top indicates the current view is 'Table: instructors'.

instructor_id	instructor_name	deparment_id	room_no
101	Dr. Rahman	1	NULL
102	Dr. Akif	2	NULL
103	Prof. Dr. Khandakar Nadim Parvez	3	NULL

Question 5. Create a temporary table named lab_test and then drop it.

Code:

```
CREATE TEMPORARY TABLE lab_test (
    id INT,
    name VARCHAR(50)
);
```

```
DROP TEMPORARY TABLE lab_test;
```

Output:

The screenshot shows the phpMyAdmin interface with the 'university_db' database selected. The left sidebar shows the database structure, including tables like classrooms, courses, departments, enrollments, instructors, and students. The main panel displays the execution of two SQL queries:

Query 1: `CREATE TEMPORARY TABLE lab_test ( id INT, name VARCHAR(50) );`

Query 2: `DROP TEMPORARY TABLE lab_test;`

Both queries returned an empty result set (i.e. zero rows) and took 0.0003 seconds to execute. The interface also includes a 'Show query box' button and links for 'Edit inline', 'Edit', and 'Create PHP code' for each query.

Question 6. Insert three new students into students.

***Before inserting new student -**

student_id	student_name	email	department_id	age	cgpa
1001	Rahim Ahmed	rahim@gmail.com	1	21	3.75
1002	Karim Hasan	karim@gmail.com	1	22	3.50
1003	Nusrat Jahan	nusrat@gmail.com	2	21	3.90
1004	Sadia Islam	sadia@gmail.com	3	23	3.25
1005	Tanvir Hossain	tanvir@gmail.com	1	2	3.60
1006	Mim Akter	mim@gmail.com	2	20	3.80

Code:

```
INSERT INTO students VALUES  
(1007,'Luckman Rafique','luckman@gmail.com',2,20,3.00),  
(1008,'Shimul Islam','shimul@gmail.com',5,24,3.82),  
(1009,'Shah Rokan','rokan@gmail.com',4,24,3.94);
```

Output:

The screenshot displays the phpMyAdmin web interface. On the left, a sidebar shows the database structure with 'university_db' expanded, listing tables like classrooms, courses, departments, enrollments, instructors, and students. The main panel shows the 'students' table selected. A green message box indicates '3 rows inserted. (Query took 0.0017 seconds.)'. Below this, the executed SQL query is shown: `INSERT INTO students VALUES (1007,'Luckman Rafique','luckman@gmail.com',2,20,3.00), (1008,'Shimul Islam','shimul@gmail.com',5,24,3.82), (1009,'Shah Rokan','rokan@gmail.com',4,24,3.94);`. At the bottom of the query box, there are links for '[Edit inline]', '[Edit]', and '[Create PHP code]'.

student_id	student_name	email	department_id	age	cgpa
1001	Rahim Ahmed	rahim@gmail.com	1	21	3.75
1002	Karim Hasan	karim@gmail.com	1	22	3.50
1003	Nusrat Jahan	nusrat@gmail.com	2	21	3.90
1004	Sadia Islam	sadia@gmail.com	3	23	3.25
1005	Tanvir Hossain	tanvir@gmail.com	1	2	3.60
1006	Mim Akter	mim@gmail.com	2	20	3.80
1007	Luckman Rafique	luckman@gmail.com	2	20	3.00
1008	Shimul Islam	shimul@gmail.com	5	24	3.82
1009	Shah Rokan	rokan@gmail.com	4	24	3.94

Question 7. Insert two new courses into courses.

***Before inserting two new courses -**

course_id	course_name	credit	department_id	semester	instructor_id
BBA101	Principle of Management	3	3	1	103
CSE101	Structured Programming	3	1	1	101
CSE201	Database System	3	1	3	101
CSE301	Computer Networks	3	1	5	101
EEE101	Basic Electrical Engineering	3	2	1	102

Code:

```
INSERT INTO courses VALUES
('MAT211','Differential Calculus',3,4,2,102),
('ENG101','English Language',4,1,1,103);
```

Output:



The screenshot shows the phpMyAdmin web interface. The left sidebar displays a database structure tree with 'university_db' selected, showing tables like classrooms, courses, departments, enrollments, instructors, and students. The main panel shows the 'courses' table selected. A yellow message box indicates '2 rows inserted. (Query took 0.0005 seconds.)'. Below this, the SQL query is displayed: `INSERT INTO courses VALUES ('MAT211','Differential Calculus',3,4,2,102), ('ENG101','English Language',4,1,1,103);`. Links for '[Edit inline]', '[Edit]', and '[Create PHP code]' are visible.

course_id	course_name	credit	department_id	semester	instructor_id
BBA101	Principle of Management	3	3	1	103
CSE101	Structured Programming	3	1	1	101
CSE201	Database System	3	1	3	101
CSE301	Computer Networks	3	1	5	101
EEE101	Basic Electrical Engineering	3	2	1	102
ENG101	English Language	4	1	1	103
MAT211	Differential Calculus	3	4	2	102

Question 8. Delete one student using student_id.

Code:

```
DELETE FROM students  
WHERE student_id=1002;
```

Output:

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'university_db' selected. The main panel shows the 'students' table. A yellow message box states '1 row deleted. (Query took 0.0002 seconds.)'. Below it, the SQL query 'DELETE FROM students WHERE student_id=1002;' is displayed. At the bottom right of the query box are links: '[Edit inline] [Edit] [Create PHP code]'.

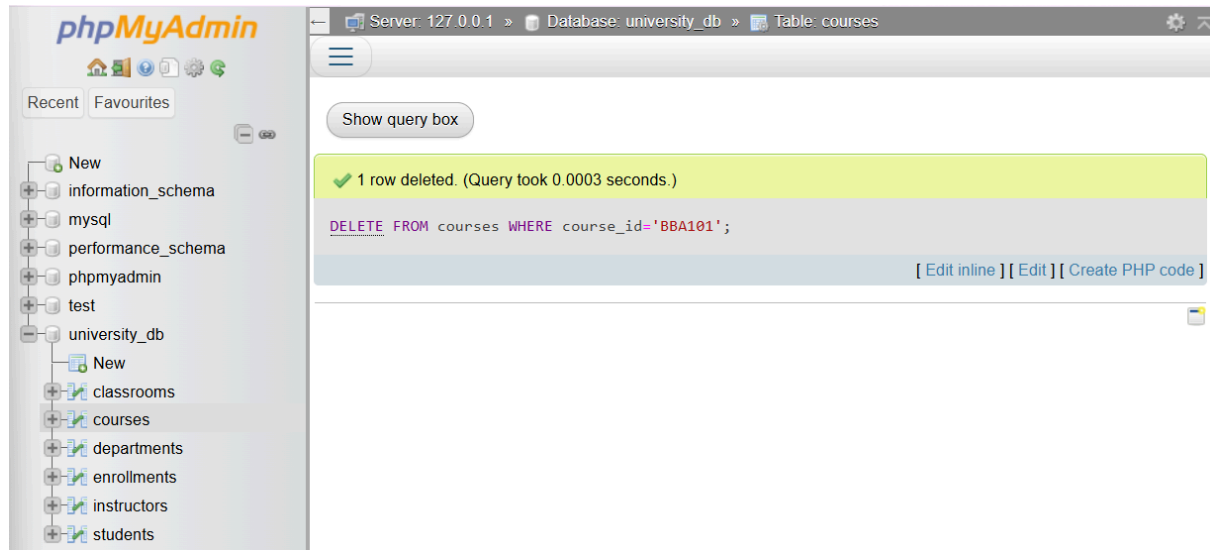
student_id	student_name	email	department_id	age	cgpa
1001	Rahim Ahmed	rahim@gmail.com	1	21	3.75
1003	Nusrat Jahan	nusrat@gmail.com	2	21	3.90
1004	Sadia Islam	sadia@gmail.com	3	23	3.25
1005	Tanvir Hossain	tanvir@gmail.com	1	2	3.60
1006	Mim Akter	mim@gmail.com	2	20	3.80
1007	Luckman Rafique	luckman@gmail.com	2	20	3.00
1008	Shimul Islam	shimul@gmail.com	5	24	3.82
1009	Shah Roka	rokan@gmail.com	4	24	3.94

Question 9. Delete one course using course_id.

Code:

```
DELETE FROM courses
WHERE course_id='BBA101';
```

Output:



The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'university_db' expanded, showing tables like 'classrooms', 'courses', 'departments', 'enrollments', 'instructors', and 'students'. The main panel shows the 'Table: courses' view. A green message bar indicates '1 row deleted. (Query took 0.0003 seconds.)'. Below this, the executed SQL query is displayed: `DELETE FROM courses WHERE course_id='BBA101';`. Links for '[Edit inline]', '[Edit]', and '[Create PHP code]' are visible.

course_id	course_name	credit	department_id	semester	instructor_id
CSE101	Structured Programming	3	1	1	101
CSE201	Database System	3	1	3	101
CSE301	Computer Networks	3	1	5	101
EEE101	Basic Electrical Engineering	3	2	1	102
ENG101	English Language	4	1	1	103
MAT211	Differential Calculus	3	4	2	102

Question 10. Display the structure of all tables using DESCRIBE.

Code:

```
DESCRIBE students;  
DESCRIBE instructors;  
DESCRIBE enrollments;  
DESCRIBE departments;  
DESCRIBE courses;  
DESCRIBE classrooms;
```

Output:

*students -

Field	Type	Null	Key	Default	Extra
student_id	int(11)	NO	PRI	NULL	
student_name	varchar(100)	NO		NULL	
email	varchar(100)	YES		NULL	
department_id	int(11)	YES		NULL	
age	int(11)	YES		NULL	
cgpa	decimal(3,2)	YES		NULL	

*instructors -

Field	Type	Null	Key	Default	Extra
instructor_id	int(11)	NO	PRI	NULL	
instructor_name	varchar(100)	NO		NULL	
deparment_id	int(11)	YES		NULL	
room_no	varchar(30)	YES		NULL	

*enrollments -

Field	Type	Null	Key	Default	Extra
enrollment_id	int(11)	NO	PRI	NULL	
student_id	int(11)	YES		NULL	
course_id	varchar(10)	YES		NULL	
marks	decimal(5,2)	YES		NULL	
semester	int(11)	YES		NULL	

*departments

Field	Type	Null	Key	Default	Extra
department_id	int(11)	NO	PRI	NULL	
department_name	varchar(50)	NO		NULL	

***courses**

Field	Type	Null	Key	Default	Extra
course_id	varchar(10)	NO	PRI	NULL	
course_name	varchar(100)	NO		NULL	
credit	int(11)	YES		NULL	
department_id	int(11)	YES		NULL	
semester	int(11)	YES		NULL	
instructor_id	int(11)	YES		NULL	

***classrooms**

Field	Type	Null	Key	Default	Extra
classroom_id	int(11)	NO	PRI	NULL	
room_number	varchar(20)	YES		NULL	
building	varchar(50)	YES		NULL	
capacity	int(11)	YES		NULL	

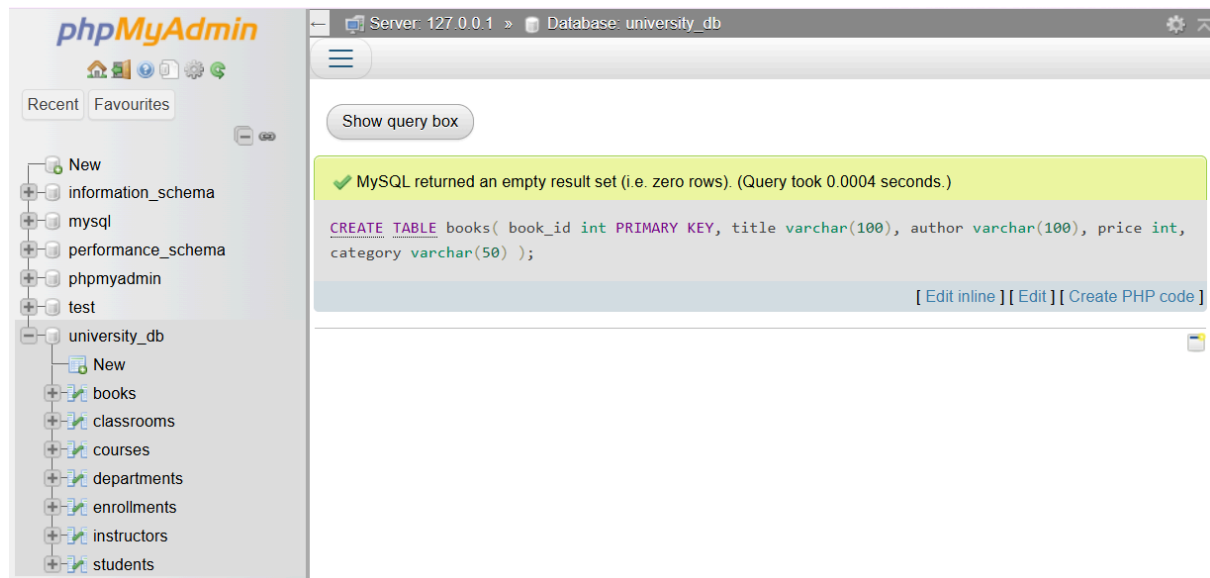
Part (ii) — Individual Practice/ Evaluation —

Question 1. Create a table named books with book_id, title, author, price, and category.

Code:

```
CREATE TABLE books(  
    book_id int PRIMARY KEY,  
    title varchar(100),  
    author varchar(100),  
    price int,  
    category varchar(50)  
);
```

Output:

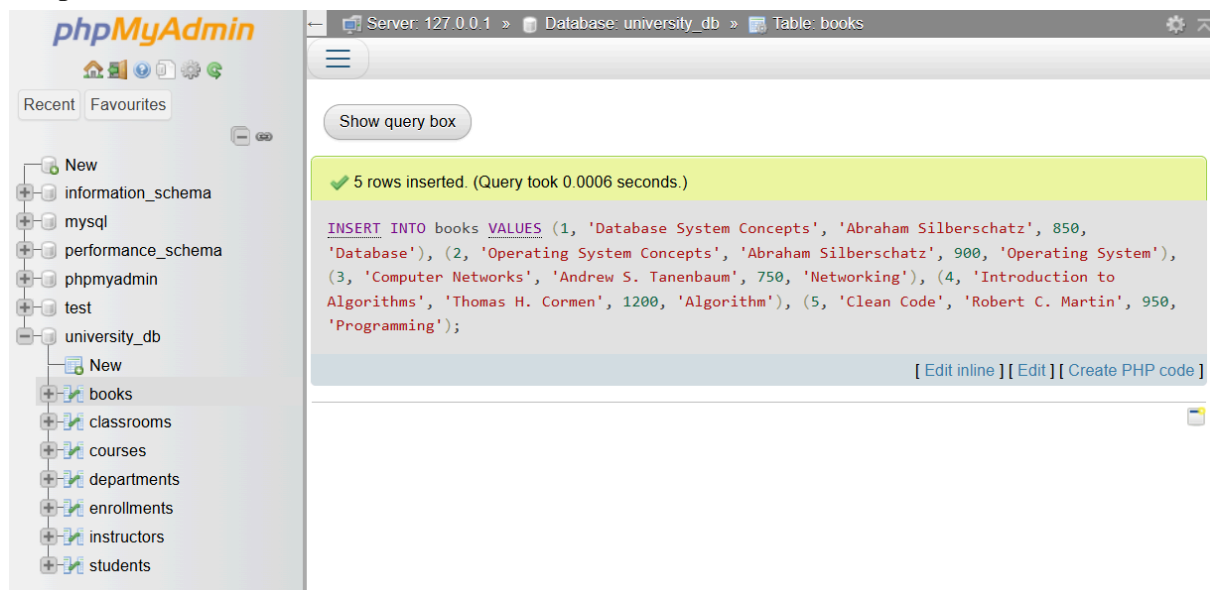


Question 2. Insert at least five records into books.

Code:

```
INSERT INTO books VALUES  
(1, 'Database System Concepts', 'Abraham Silberschatz', 850, 'Database'),  
(2, 'Operating System Concepts', 'Abraham Silberschatz', 900, 'Operating System'),  
(3, 'Computer Networks', 'Andrew S. Tanenbaum', 750, 'Networking'),  
(4, 'Introduction to Algorithms', 'Thomas H. Cormen', 1200, 'Algorithm'),  
(5, 'Clean Code', 'Robert C. Martin', 950, 'Programming');
```

Output:



The screenshot shows the phpMyAdmin web interface. On the left is a sidebar with a database tree structure. The main panel on the right shows the 'Table: books' view. A green message box at the top of the main panel states: '5 rows inserted. (Query took 0.0006 seconds.)'. Below this, the SQL query is displayed: `INSERT INTO books VALUES (1, 'Database System Concepts', 'Abraham Silberschatz', 850, 'Database'), (2, 'Operating System Concepts', 'Abraham Silberschatz', 900, 'Operating System'), (3, 'Computer Networks', 'Andrew S. Tanenbaum', 750, 'Networking'), (4, 'Introduction to Algorithms', 'Thomas H. Cormen', 1200, 'Algorithm'), (5, 'Clean Code', 'Robert C. Martin', 950, 'Programming');`. At the bottom of the query box are links: [Edit inline] [Edit] [Create PHP code].

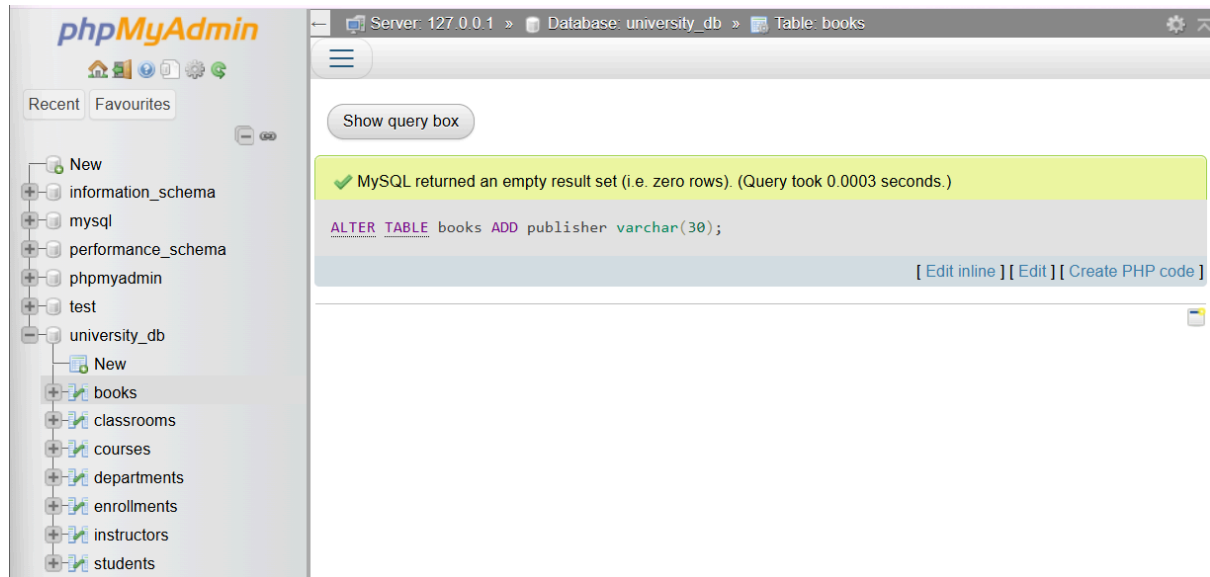
book_id	title	author	price	category
1	Database System Concepts	Abraham Silberschatz	850	Database
2	Operating System Concepts	Abraham Silberschatz	900	Operating System
3	Computer Networks	Andrew S. Tanenbaum	750	Networking
4	Introduction to Algorithms	Thomas H. Cormen	1200	Algorithm
5	Clean Code	Robert C. Martin	950	Programming

Question 3. Add a publisher column to books.

Code:

```
ALTER TABLE books  
ADD publisher varchar(30);
```

Output:



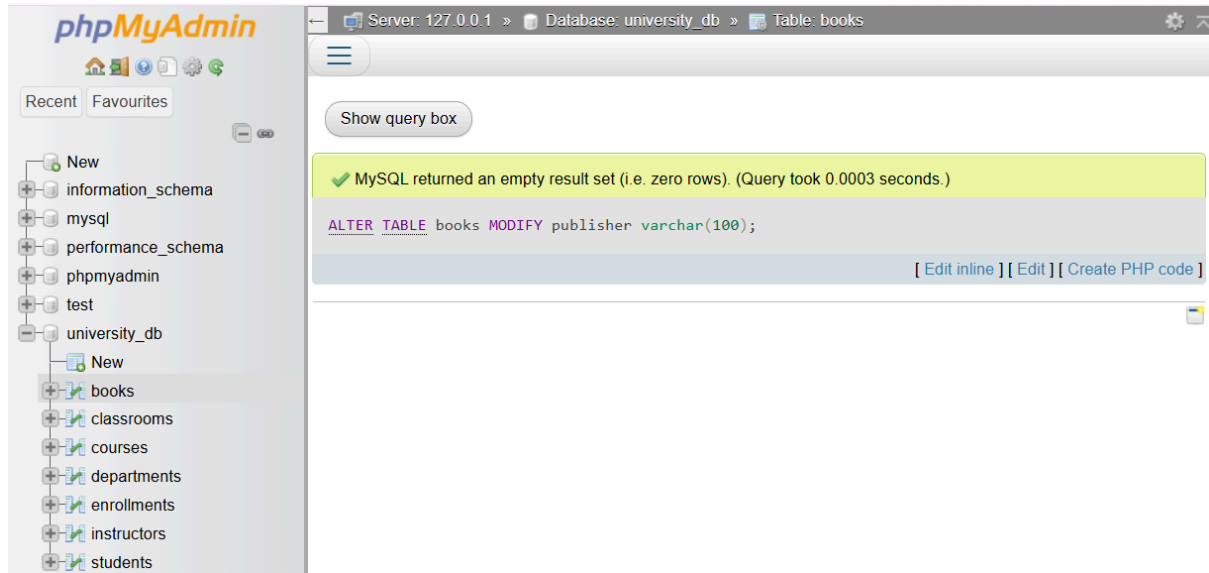
book_id	title	author	price	category	publisher
1	Database System Concepts	Abraham Silberschatz	850	Database	NULL
2	Operating System Concepts	Abraham Silberschatz	900	Operating System	NULL
3	Computer Networks	Andrew S. Tanenbaum	750	Networking	NULL
4	Introduction to Algorithms	Thomas H. Cormen	1200	Algorithm	NULL
5	Clean Code	Robert C. Martin	950	Programming	NULL

Question 4. Modify publisher to VARCHAR(100).

Code:

```
ALTER TABLE books  
MODIFY publisher varchar(100);
```

Output:

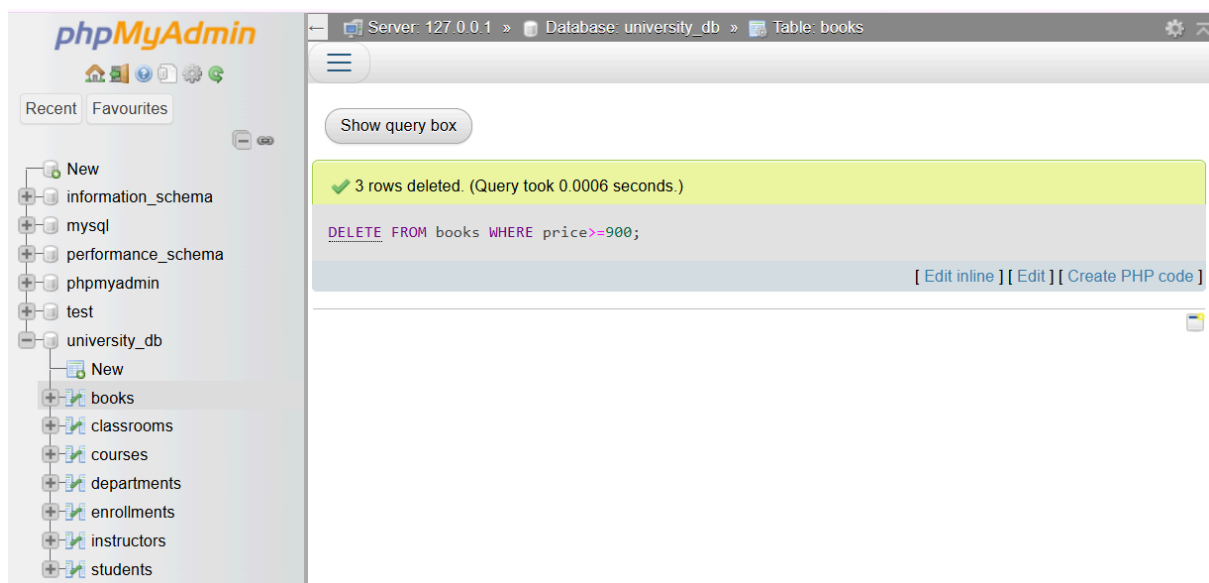


Question 5. Delete one book using a condition.

Code:

```
DELETE FROM books  
WHERE price >= 900;
```

Output:



book_id	title	author	price	category	publisher
1	Database System Concepts	Abraham Silberschatz	850	Database	NULL
3	Computer Networks	Andrew S. Tanenbaum	750	Networking	NULL

Question 6. Create a temporary table and remove it with DROP TABLE.

Code:

```
CREATE TEMPORARY TABLE temp_books (
    book_id INT,
    title VARCHAR(100)
);
```

```
DROP TABLE temp_books;
```

Output:

The screenshot shows the phpMyAdmin interface with the 'university_db' database selected. The left sidebar shows the database structure, including tables like 'books', 'classrooms', 'courses', 'departments', 'enrollments', 'instructors', and 'students'. The main panel displays the execution of two SQL queries:

- The first query is `CREATE TEMPORARY TABLE temp_books ( book_id INT, title VARCHAR(100) );`. The output shows a green checkmark and the message: "MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)". Below the query, there are links for "[Edit inline]", "[Edit]", and "[Create PHP code]".
- The second query is `DROP TABLE temp_books;`. The output also shows a green checkmark and the message: "MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)". Below the query, there are links for "[Edit inline]", "[Edit]", and "[Create PHP code]".

Question 7. Explain the difference between CREATE, ALTER, DROP, INSERT, and DELETE.

Difference between CREATE, ALTER, DROP, INSERT and DELETE -

SQL Command	Purpose	Example
CREATE	Used to create new database object, such as a table.	CREATE TABLE students(student_id int PRIMARY KEY, student_name varchar(100) NOT NULL, email varchar(100), department_id int, age int, cgpa decimal(3,2));
ALTER	Used to modify the structure of an existing table, such as adding or changing a column.	ALTER TABLE students ADD email VARCHAR(30);
DROP	Used to permanently remove database object, such as a table.	DROP TABLE students;
INSERT	Used to add new records/rows or values to a table.	INSERT INTO students VALUES (1001,'Rahim Ahmed','rahim@gmail.com',1,21,3.75);
DELETE	Used to remove existing records/rows/values from a table.	DELETE FROM students WHERE student_id=1001;

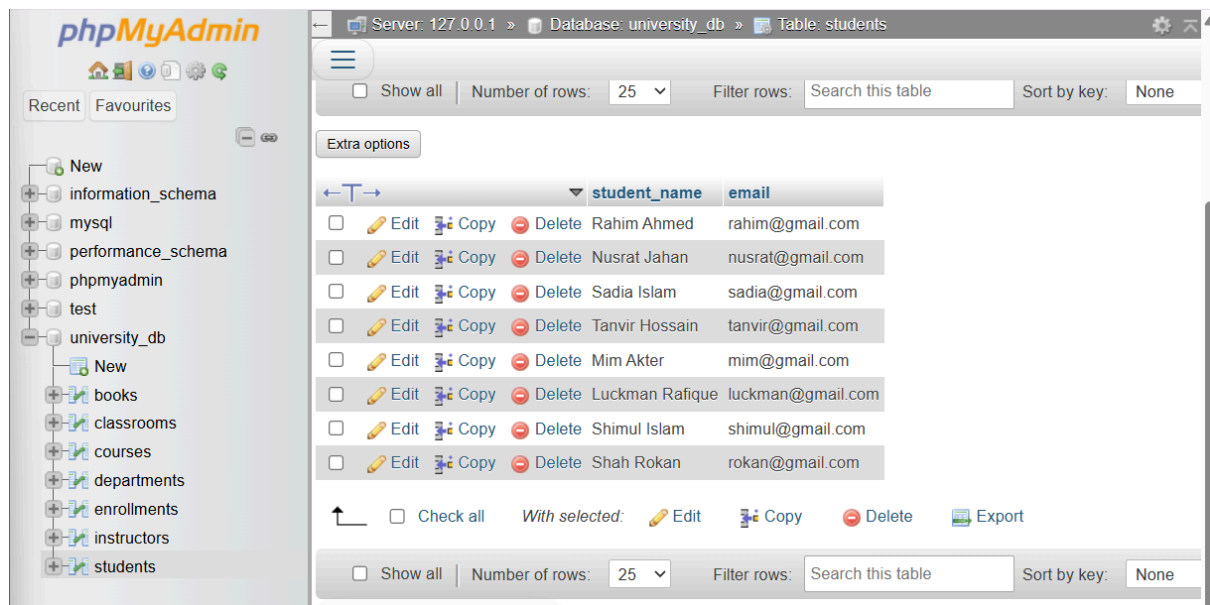
Part (iii) — Practice Problem —

Question 1. Write an SQL query to display the student name and email address of all students.

Code:

```
SELECT student_name, email  
FROM students;
```

Output:



The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a tree view of databases and tables. The 'university_db' database is selected, and the 'students' table is highlighted. The main panel displays the 'students' table with columns 'student_name' and 'email'. The table contains 10 rows of data. At the top of the main panel, there is a navigation bar with 'Server: 127.0.0.1', 'Database: university_db', and 'Table: students'. Below this is a toolbar with options like 'Show all', 'Number of rows: 25', 'Filter rows: Search this table', and 'Sort by key: None'. Below the toolbar is an 'Extra options' button. Below that is a table with 10 rows. Each row has a checkbox, an 'Edit' button, a 'Copy' button, and a 'Delete' button, followed by the 'student_name' and 'email' values. At the bottom of the table is a 'Check all' checkbox and a 'With selected:' section with 'Edit', 'Copy', 'Delete', and 'Export' buttons. Below the table is another toolbar with 'Show all', 'Number of rows: 25', 'Filter rows: Search this table', and 'Sort by key: None'.

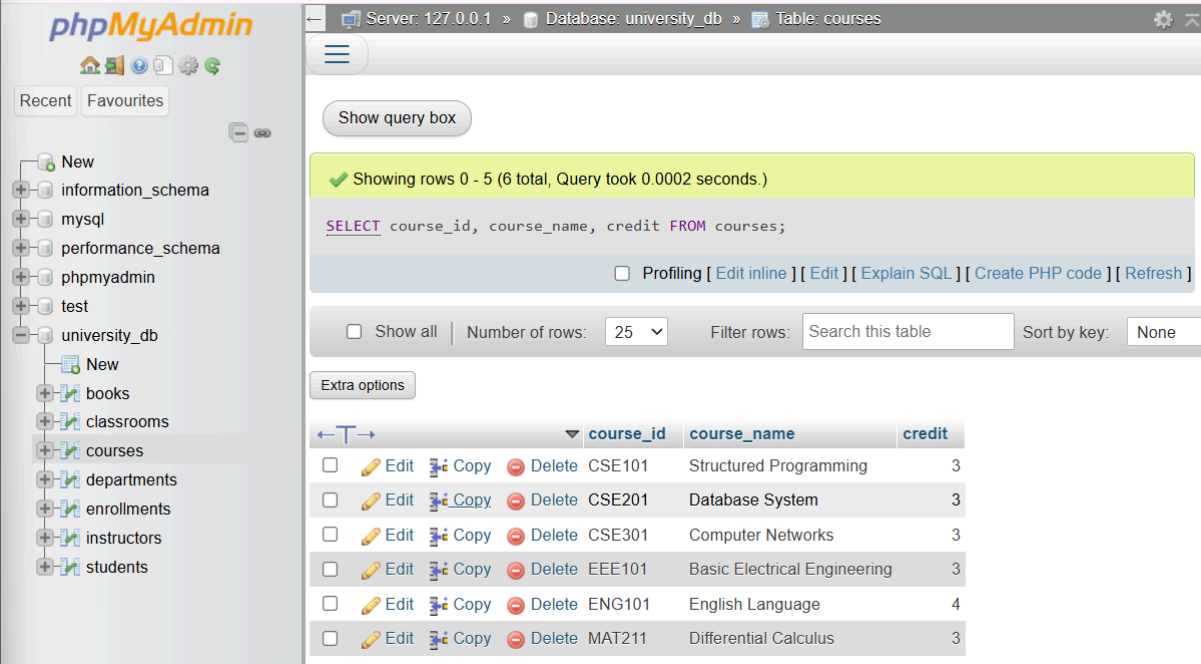
	student_name	email
☐	Rahim Ahmed	rahim@gmail.com
☐	Nusrat Jahan	nusrat@gmail.com
☐	Sadia Islam	sadia@gmail.com
☐	Tanvir Hossain	tanvir@gmail.com
☐	Mim Akter	mim@gmail.com
☐	Luckman Rafique	luckman@gmail.com
☐	Shimul Islam	shimul@gmail.com
☐	Shah Roka	rokan@gmail.com

Question 2. Write an SQL query to display the course ID, course name, and credit of all courses.

Code:

```
SELECT course_id, course_name, credit
FROM courses;
```

Output:



The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a database tree. The 'university_db' database is selected, and the 'courses' table is highlighted. The main panel displays the SQL query: `SELECT course_id, course_name, credit FROM courses;`. A green status bar indicates the query was successful, showing rows 0-5 of 6 total. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. Further down, there are controls for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (a search box), and 'Sort by key' (set to None). An 'Extra options' button is also present. The results are shown in a table with columns 'course_id', 'course_name', and 'credit'.

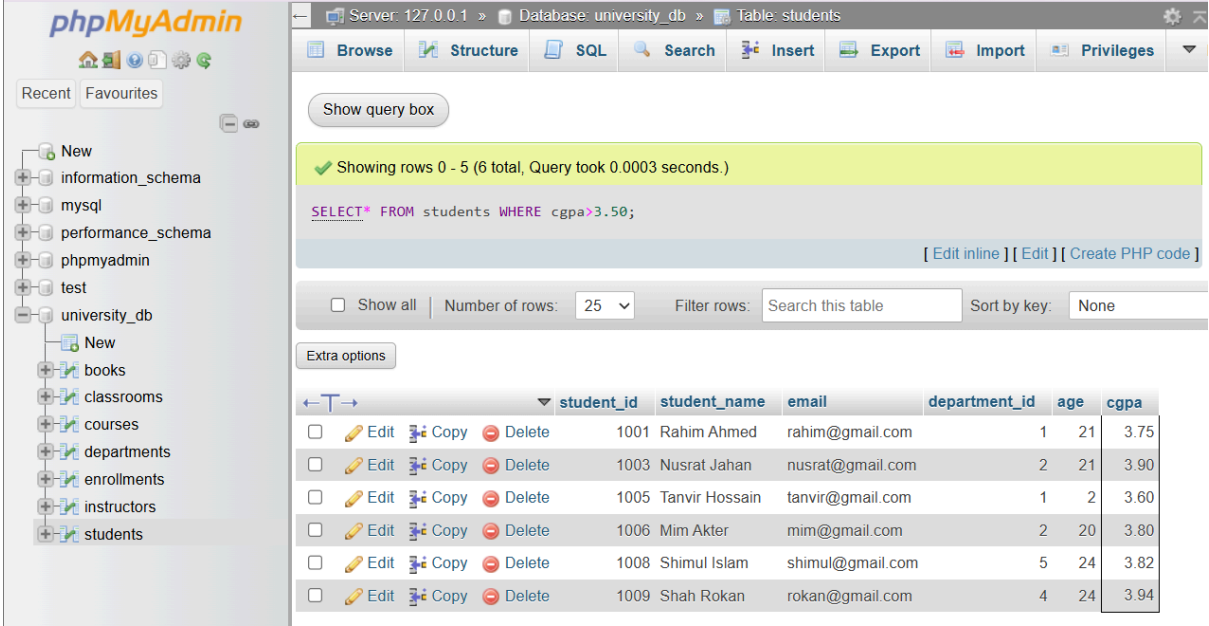
	course_id	course_name	credit
☐ Edit Copy Delete	CSE101	Structured Programming	3
☐ Edit Copy Delete	CSE201	Database System	3
☐ Edit Copy Delete	CSE301	Computer Networks	3
☐ Edit Copy Delete	EEE101	Basic Electrical Engineering	3
☐ Edit Copy Delete	ENG101	English Language	4
☐ Edit Copy Delete	MAT211	Differential Calculus	3

Question 3. Write an SQL query to display all students whose CGPA is greater than 3.50.

Code:

```
SELECT*  
FROM students  
WHERE cgpa>3.50;
```

Output:



The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a tree view of databases and tables. The main panel displays the SQL query: `SELECT* FROM students WHERE cgpa>3.50;`. Below the query, a message indicates that 6 rows are shown. The results are displayed in a table with columns: student_id, student_name, email, department_id, age, and cgpa. The table contains 6 rows of student data.

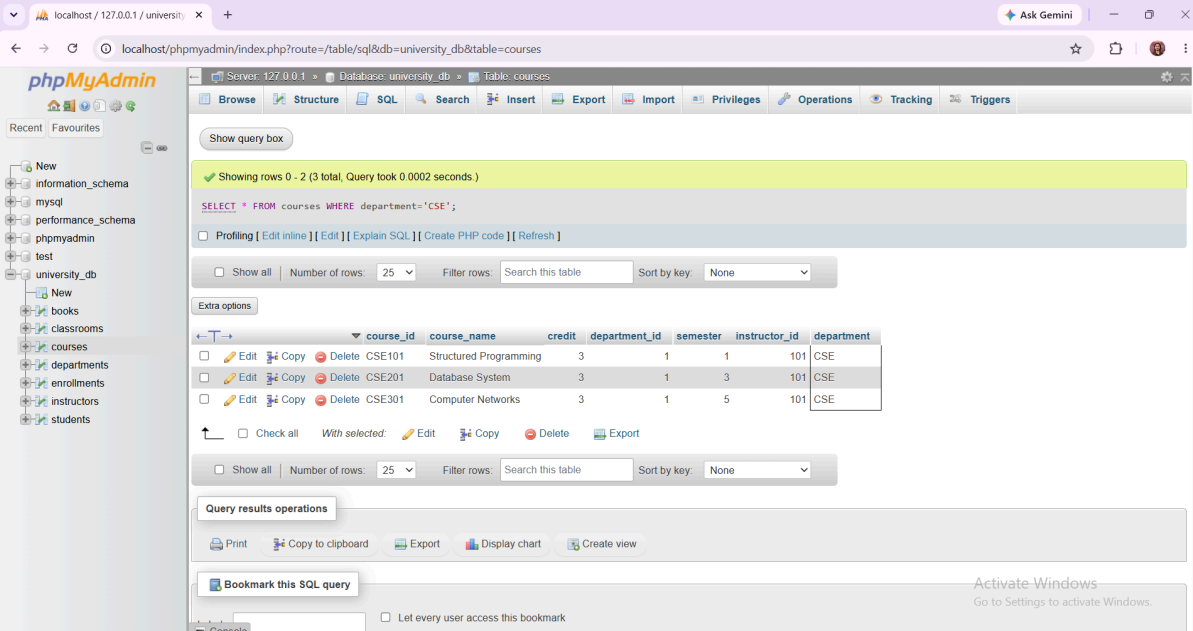
	student_id	student_name	email	department_id	age	cgpa
☐	1001	Rahim Ahmed	rahim@gmail.com	1	21	3.75
☐	1003	Nusrat Jahan	nusrat@gmail.com	2	21	3.90
☐	1005	Tanvir Hossain	tanvir@gmail.com	1	2	3.60
☐	1006	Mim Akter	mim@gmail.com	2	20	3.80
☐	1008	Shimul Islam	shimul@gmail.com	5	24	3.82
☐	1009	Shah Rokan	rokan@gmail.com	4	24	3.94

Question 4. Write an SQL query to display all courses that belong to the CSE department.

Code:

```
SELECT *  
FROM courses  
WHERE department='CSE';
```

Output:



The screenshot shows the phpMyAdmin interface for a database named 'university_db'. The 'courses' table is selected, and the SQL query `SELECT * FROM courses WHERE department='CSE';` has been executed. The results show 3 rows of data for CSE courses.

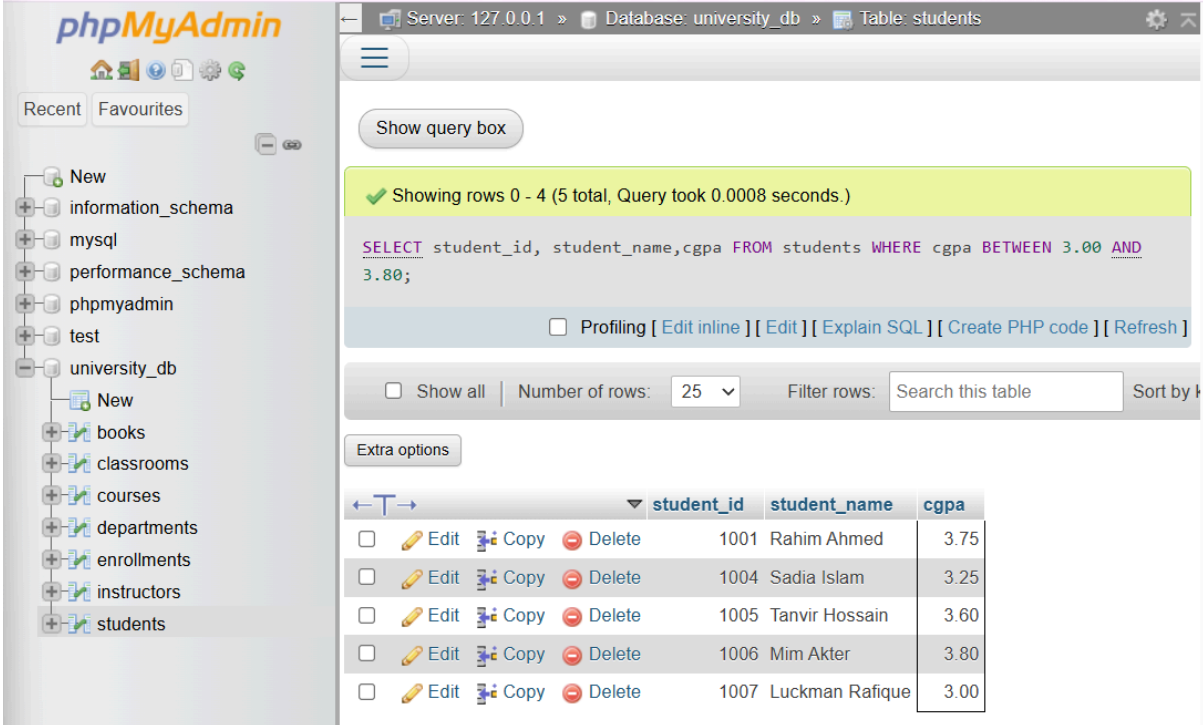
course_id	course_name	credit	department_id	semester	instructor_id	department
CSE101	Structured Programming	3	1	1	101	CSE
CSE201	Database System	3	1	3	101	CSE
CSE301	Computer Networks	3	1	5	101	CSE

Question 5. Write an SQL query to display the student ID, student name, and CGPA of students whose CGPA is between 3.00 and 3.80.

Code:

```
SELECT student_id, student_name,cgpa
FROM students
WHERE cgpa BETWEEN 3.00 AND 3.80;
```

Output:



The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a tree view of databases and tables. The 'university_db' database is selected, and the 'students' table is highlighted. The main panel displays the SQL query: `SELECT student_id, student_name,cgpa FROM students WHERE cgpa BETWEEN 3.00 AND 3.80;`. A green status bar indicates 'Showing rows 0 - 4 (5 total, Query took 0.0008 seconds.)'. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A table of results is shown with columns 'student_id', 'student_name', and 'cgpa'. The table contains 5 rows of data. Each row has checkboxes and links for 'Edit', 'Copy', and 'Delete'.

	student_id	student_name	cgpa
☐ Edit Copy Delete	1001	Rahim Ahmed	3.75
☐ Edit Copy Delete	1004	Sadia Islam	3.25
☐ Edit Copy Delete	1005	Tanvir Hossain	3.60
☐ Edit Copy Delete	1006	Mim Akter	3.80
☐ Edit Copy Delete	1007	Luckman Rafique	3.00